

Technical Notes: Reachability-Constrained Riemannian Planning

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I. SCOPE OF THESE NOTES

These notes accompany the planning manuscript for author review and reproducibility. They are not a claim that an additional PDF appendix is permitted by the IROS 2027 submission rules. The main paper is self-contained; current venue instructions must be checked before submission. The project is separate from controller training and grasp acquisition.

II. DIFFERENTIATING A SECURED GRASP

Write the object transform as $T = (R, p)$ and a fixed grasp as $A = (R_A, a)$. The world palm target is $(RR_A, p + Ra)$. For a world angular velocity ω , $\dot{R} = [\omega]_{\times} R$ and

$$\frac{d}{dt}(p + Ra) = u + [\omega]_{\times} Ra = \begin{bmatrix} I & -[Ra]_{\times} \\ & \omega \end{bmatrix} \begin{bmatrix} u \\ \omega \end{bmatrix}. \quad (1)$$

Weighting these velocities by $W \succeq 0$, summing over the four hands, and adding a positive baseline gives

$$v^{\top} G(T) v = v^{\top} B v + \sum_h \|W^{1/2} J_h(T) v\|^2. \quad (2)$$

This directly establishes positive definiteness and $G(T) \succeq B$. It also states precisely what the objective values: palm translation under a chosen anisotropic penalty. It does not measure actuator effort, potential energy, collision clearance, or distance to an IK singularity. Those considerations enter separate constraints or local posture refinement.

III. WHY THE INFORMED TEST RESPECTS ROTATION TOPOLOGY

An orientation can be represented by either quaternion q or $-q$. Euler angles have additional chart identifications. Neither representation therefore permits a globally valid Euclidean displacement without accounting for those identifications. The principal angle

$$\theta(R_b R_a^{\top}) = \cos^{-1} \left(\frac{\text{tr}(R_b R_a^{\top}) - 1}{2} \right) \quad (3)$$

is independent of quaternion sign. Numerically, the implementation uses the norm of a principal rotation vector rather than evaluating this trace formula near its conditioning limits.

Under $B = \text{diag}(I, \rho^2 I)$, translation and orientation form a Riemannian product. For any path, its translation length bounds the endpoint displacement and its rotational length bounds the principal rotation angle. Minkowski's integral inequality then yields

$$\int \sqrt{\|u\|^2 + \rho^2 \|\omega\|^2} dt \geq \sqrt{\|p_b - p_a\|^2 + \rho^2 \theta^2}. \quad (4)$$

Straight translation and a constant-speed principal rotation attain the product distance with synchronized parameterization. Combining this with $G \succeq B$ proves the main paper's bound.

Applying the bound to both portions of a path through a pose T gives the informed test.

The midpoint quadrature used for graph edges is not known to overestimate the true G -length. Consequently, an incumbent obtained solely from that quadrature cannot turn the implementation into an exact admissible optimal-search algorithm. The code uses the bound as a finite-budget sampling heuristic and makes no asymptotic optimality claim. This distinction matters even though the geometric distance bound itself is exact.

IV. GLOBAL DISTANCE-TIME INCUMBENT

For an audited route with dense object poses T_k and witness configurations q_k , define

$$D = \sum_k \|p_{k+1} - p_k\|, \quad T = \mathcal{R}(\{T_k, q_k\}), \quad \Phi = \frac{1}{2} D / D_{\text{ref}} + \frac{1}{2} T / T_{\text{ref}}, \quad (5)$$

where $\mathcal{R}$ is the shared rate and sampled-acceleration retimer described later. The reference pair $(D_{\text{ref}}, T_{\text{ref}})$ is frozen after the first independently audited candidate in each finite-budget run. This makes the combined score dimensionless while preserving a fixed ordering during that run. It is a route-quality objective, not an energy or torque objective.

The roadmap continues to use the grasp-induced metric for nearest-neighbor construction and its product lower bound for informed sampling. Whenever Dijkstra produces a fully validated candidate, the implementation evaluates Φ on the complete dense route and replaces the incumbent only if the score decreases. Thus the geometric graph cost and the selection objective are deliberately separated. The method does not claim that the first candidate is a lower bound, and it does not claim global optimality.

The visibility postprocessor constructs a candidate chord between two existing knots, uses the $SE(3)$ retraction and bounded IK projection at interior samples, and retains the two original endpoint configurations. Let γ be the current route and $\tilde{\gamma}$ an independently audited chord replacement. The acceptance rule is

$$\tilde{\gamma} \leftarrow \gamma \iff \Phi(\tilde{\gamma}) < \Phi(\gamma) - \varepsilon, \quad \varepsilon = 10^{-7}. \quad (6)$$

Consequently, by induction over accepted replacements, the returned route has no larger measured Φ than the initial audited route. This statement is only about the finite set of sampled candidates and the implemented retimer; independent closure, limit and collision audits are still required after the final replacement.

V. CONSTRAINT LINEARIZATION AND ACTIVE LIMITS

The orientation residual is $e_R = \text{Log}(R_{\text{palm}} R_{\text{desired}}^{\top})^{\vee}$. Away from zero residual, its exact differential contains an

inverse left Jacobian of $SO(3)$. The implementation uses the geometric palm angular Jacobian as a local approximation, weighted by 0.25 m, and re-evaluates the nonlinear residual after every update. At closure the inverse left Jacobian is the identity. Numerical acceptance is based on the nonlinear residual, not on assuming that the linearized subproblem was exact at large error.

For each bounded scalar joint, a step must satisfy $q_{\min} - q \leq d \leq q_{\max} - q$. The solver intersects these intervals with a 0.12 trust box. Base coordinates have only the trust-box bounds within IK; their floor height and inclination are not variables. The bounded least-squares problem is solved with SciPy’s active-set BVLS method, rather than applying an unconstrained step and depending on clipping to resolve active limits. A final numerical clipping guard protects against roundoff.

The two lift-equality rows ensure equal extension of each robot’s paired stages. A nominal node contains 44 variable coordinates and 26 residuals, leaving an 18-dimensional local posture freedom only where the equality Jacobian has full row rank. Active bounds or singularities can invalidate this count. We do not replace these conditions with a global smooth-manifold assumption.

VI. STORED CONFIGURATIONS AND PATH CONTINUITY

Consider two accepted graph edges incident at vertex v . If the first connection ends at (T_v, q_v) and the next starts at that same pair, their continuous local interpolants can be concatenated without a configuration jump. If only T_v is shared, distinct IK branches $q_v^{(1)}$ and $q_v^{(2)}$ can create a discontinuity even though both satisfy the same grasp constraints. Storing the full configuration at each vertex resolves this endpoint ambiguity.

This observation does not prove that every interpolated state satisfies the constraints. The implementation samples projected connections, then checks midpoints. The separate export audit evaluates the normalized-quaternion interpolation actually used by the position executor at fractions 1/4, 1/2, and 3/4. The maximum observed scalar-joint increment in the original six-scene refinement is 0.0486 in the joint’s units; the 0.045 value controls initial subdivision from endpoint differences and is not a certified post-projection bound. All such checks remain discrete.

VII. POSTURE DESCENT DERIVATION

For a local equality Jacobian J and $H \succ 0$, minimize $g^\top v + \frac{1}{2}v^\top H v$ subject to $Jv = 0$. Stationarity gives $v = -H^{-1}(g + J^\top \mu)$. Enforcing the constraint gives

$$\mu = -(JH^{-1}J^\top)^{-1}JH^{-1}g. \quad (7)$$

Substitution yields the metric-orthogonal tangent step in the main manuscript. Multiplication of stationarity by $v^\top$ gives $g^\top v = -v^\top H v$. The implementation adds regularization to avoid singular inverses; exact orthogonality and the exact descent identity therefore need not hold numerically. It instead validates the reprojected result and independently audits the entire exported path, reverting to the audited roadmap path if required.

VIII. EXECUTION AND REPRODUCIBILITY BOUNDARIES

The exported palm references compose the interpolated object pose with each fixed grasp transform. This composition preserves relative object-to-palm transforms exactly. The robot positions are a separate, constrained realization of those references. Base pose targets feed a swerve feedback controller because the wheel actuators have velocity semantics. Position targets are also augmented by gravity feedforward during execution.

The simulator initializes its state once and then advances dynamics. Permanent high-impedance welds implement the assumed secure grasps. The controller cannot establish a frictional grasp-retention result under this model. Similarly, soft joint-limit violations measured in dynamics are reported even when the geometric target remains strictly within its limits. The six task environments are simulated with light payloads from 1.0 to 2.0 kg; the work does not demonstrate transport of full-mass furniture.

The accompanying data distinguish six nominal plans, twelve additional seeded planning runs, 36 paired metric trials, 432 isolated pose probes, six whole-body dynamics rollouts and six frozen-policy rollouts. Repeated seeds on a fixed scene set are not independent evidence of population-wide generalization. Timing was collected on a shared workstation. The manuscript therefore reports descriptive paired outcomes without population-level significance claims.

IX. JOINT DISTANCE AND TIME REFINEMENT

The long-route initializer connects room centers, which can induce rotation at zero translation. Its geometric objective cannot distinguish timing schedules. We retain its feasible route as a candidate but replace each room-center turn with a family of quadratic curves

$$p_{xy}(u) = (1-u)^2a + 2u(1-u)c + u^2b, \quad u \in [0, 1], \quad (8)$$

where c is the room center and a, b are cut points on the incoming and outgoing paths. The broadside formation yaw follows the curve tangent plus $\pi/2$; residual endpoint tilt is interpolated on $SO(3)$. Endpoint positions, object orientations and both robot configurations are preserved exactly. Interior poses undergo bounded IK, projected edge checking and an independent closure/collision audit. We test cut radii 1, 1.5, 2, 2.5, 3 m as well as the unchanged turn. The radius is a cut distance, not a constant curvature radius. The room itinerary and corridor obstacle motions remain fixed.

For candidate path γ , let $D(\gamma) = \sum_k \|\Delta p_k\|$ and let $T(\gamma)$ be its duration under the common retimer. We minimize

$$J_\lambda(\gamma) = \lambda \frac{D(\gamma)}{D(\gamma_0)} + (1-\lambda) \frac{T(\gamma)}{T(\gamma_0)}, \quad \lambda = \frac{1}{2}. \quad (9)$$

The baseline γ_0 is retimed by the same routine before normalization. All feasible combinations are enumerated (at most $6^3 = 216$). Weight sweeps at $\lambda = 0.25, 0.5, 0.75$ reuse the identical candidate set. This is a finite, constrained refinement rather than a globally optimal kinodynamic planner.

Finite-family nonincrease. If γ_0 belongs to the audited candidate set, selecting its minimum- J_λ member gives $J_\lambda(\gamma^*) \leq 1$. This follows immediately from feasibility of γ_0

TABLE I

PAIRED SEED-0 EXECUTION EFFICIENCY ON SCENES WHERE BOTH PRIOR-ONLY LIFTING AND THE GUIDED ROADMAP COMPLETE THE TASK. VALUES ARE PERCENTAGE REDUCTIONS: HIGHER IS BETTER. POSITIVE VALUES FAVOR OURS (THE GUIDED ROADMAP). MECHANICAL WORK IS NOT ELECTRICAL ENERGY.

Scene	Travel (%) $\uparrow$	Time (%) $\uparrow$	Work (%) $\uparrow$
Gallery Panel	1.33	1.00	17.88
Workshop Beam	1.16	1.67	14.21
Living Room	1.04	4.93	21.49
Warehouse Crate	0.40	0.45	20.15

and the definition of a minimum. It does not imply that each cost separately decreases, or that any path outside the tested family is inferior.

For timing, define $\Delta s_k^2 = \|\Delta p_k\|^2 + 0.5^2 \|\Delta \theta_k\|^2$. Each segment receives a minimum duration from object translation/rotation, every scalar joint, base translation and base yaw limits. A forward/backward pass imposes $v_{k+1}^2 \leq v_k^2 + 2a_s \Delta s_k$ with $a_s = 0.04$ and zero endpoint path speed. The preliminary duration is

$$\Delta t_k = \max \left\{ \Delta t_k^{\min}, \frac{2\Delta s_k}{v_k + v_{k+1}} \right\}. \quad (10)$$

Physical accelerations, including start/stop estimates, are then estimated from neighboring segment velocities using their midpoint time separation. Limits are 0.06 m/s² for object translation, 0.15 rad/s² for rotation, 0.30 joint-units/s² per scalar joint, 0.08 m/s² for each base and 0.20 rad/s² for base yaw. Rates are 0.09 m/s, 0.065 rad/s, 0.16 joint-units/s, 0.12 m/s and 0.15 rad/s respectively.

Sampled timing feasibility. Let r be the largest ratio

of sampled physical acceleration to its bound. The implementation first increases durations locally at violating neighboring segments, rechecking all channels after each update. Uniformly scaling all segment durations by $\beta = \sqrt{\max(1, r)}$ divides segment velocities by β and their finite-difference accelerations by β^2 . Thus it preserves the rate bounds and enforces the sampled acceleration bounds. This result concerns the discrete timing model. It does not certify continuous accelerations of piecewise interpolants, actuator torque limits, dynamic stability or minimum achievable execution time. Those require separate dynamics validation. The final scaling safeguard is conservative; the local repair reduces unnecessary slowing of distant segments.

TABLE II

ALGORITHMIC TRADEOFFS. THESE PROPERTIES EXPLAIN THE BENCHMARK DESIGN; THEY ARE NOT CLAIMS OF UNIVERSAL PERFORMANCE RANKINGS.

Method	Useful property	Limitation in this task
RRT-Connect	Bidirectional first-solution search	Does not optimize completed-path cost
BIT*	Informed batch search and anytime optimization	Full-body feasibility queries consume the solve budget
AORRTC	Optimizing RRT-Connect	Native path length does not minimize execution time
Informed RRT*	Informed sampling and rewiring	Requires costly closed-chain feasibility queries
Uniform roadmap	Removes scene-prior dependence	Sparse pose samples miss narrow feasible connections
Prior-only lift	Isolates the supplied itinerary	No recovery from a rejected spatial connection
Guided roadmap	Grasp-aware cost and structured proposals	Sensitive to itinerary, proposal layers and one stored IK branch

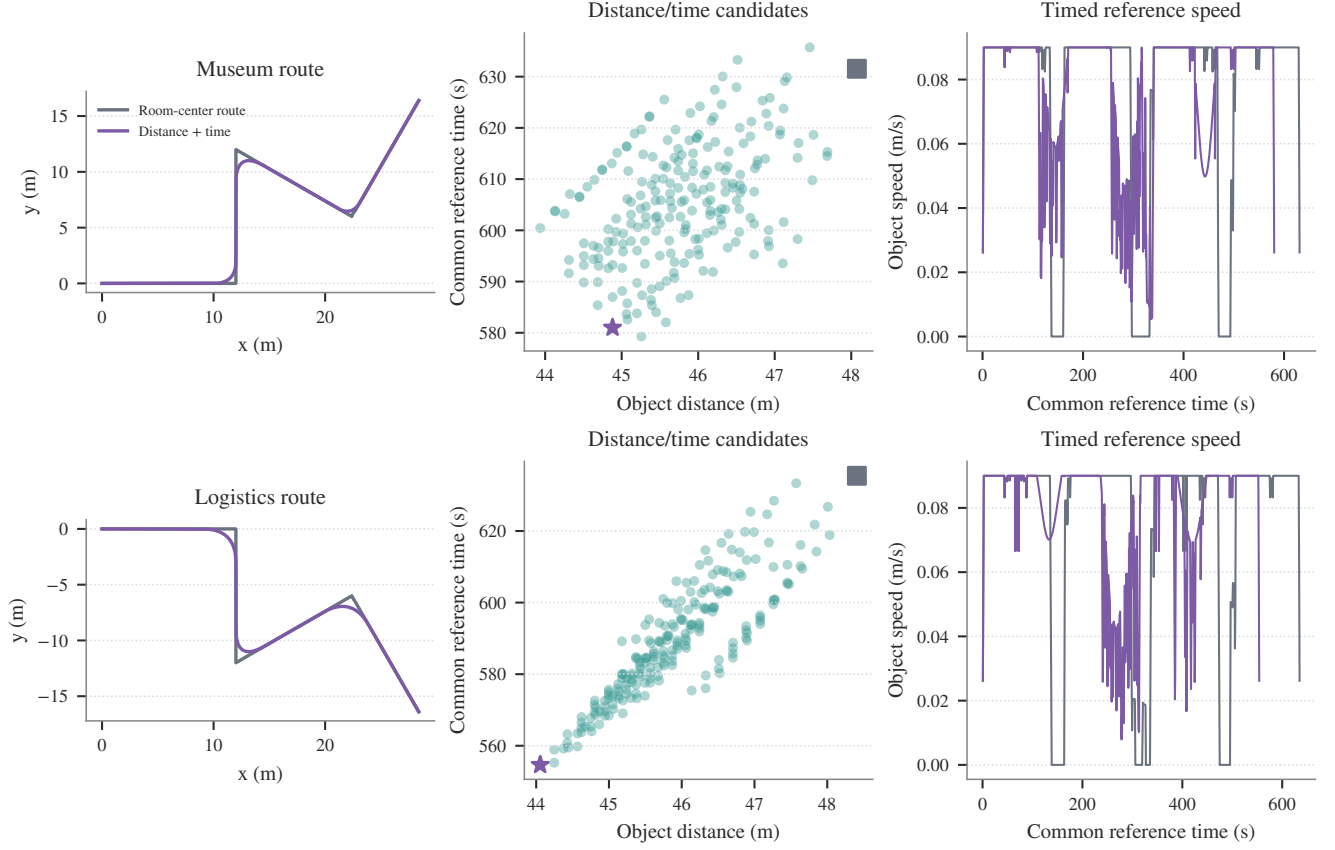


Fig. 1. Distance–time refinement in both five-room scenes. Left: unchanged and selected routes. Middle: all feasible corner combinations under common timing. Right: translational speed profiles. The selected route minimizes a normalized sum; the set is finite and itinerary-conditioned.

TABLE III

MATCHED SCENE-AND-SEED EXECUTIONS. BOTH METHODS MUST COMPLETE THE TASK. ORANGE MARKS THE LOWER VALUE WITHIN EACH PAIR, INCLUDING TIES. THESE CONDITIONAL COMPARISONS COMPLEMENT THE FULL SUCCESS/FAILURE BENCHMARK; THEY DO NOT ESTABLISH POPULATION-WIDE SUPERIORITY. WORK IS ACTUATOR MECHANICAL WORK, NOT BATTERY ENERGY.

Scene / seed	Method	Travel (m) ↓	Executed (s) ↓	Work (J) ↓
Gallery Floor Clearance / 2	AORRTC	2.21	28.98	391.97
Gallery Floor Clearance / 2	Ours	2.31	41.88	681.63
Gallery Panel / 0	Prior-only lift	6.27	96.20	1015.32
Gallery Panel / 0	Ours	6.18	95.24	833.76
Living Room / 0	Prior-only lift	7.24	118.12	851.79
Living Room / 0	Ours	7.16	112.30	668.77
Warehouse Crate / 0	Prior-only lift	7.44	107.22	963.06
Warehouse Crate / 0	Ours	7.41	106.74	769.02
Workshop Beam / 0	Prior-only lift	6.99	121.08	674.12
Workshop Beam / 0	Ours	6.91	119.06	578.34

Expanded scenes: audited successful plans / 3 seeds

atelier stair soffit	0/3	0/3
bedroom	0/3	0/3
dock case steps	0/3	0/3
foundry boss portal	0/3	0/3
joinery beam tunnel	0/3	0/3
joinery deep beam tunnel	0/3	0/3
living room painting	0/3	0/3
loading dock	0/3	0/3
loft low beam	0/3	0/3
office table	0/3	0/3
restaurant	0/3	0/3
warehouse panel	0/3	0/3
workshop glazing channel	0/3	0/3

Grasp pullout metric

Fig. 2. Expanded unchanged-scene study. Each cell counts independently audited plans over three seeds. Both metrics share the same proposals and feasibility checks. Failures expose the finite roadmap and nominal IK branch limitations.

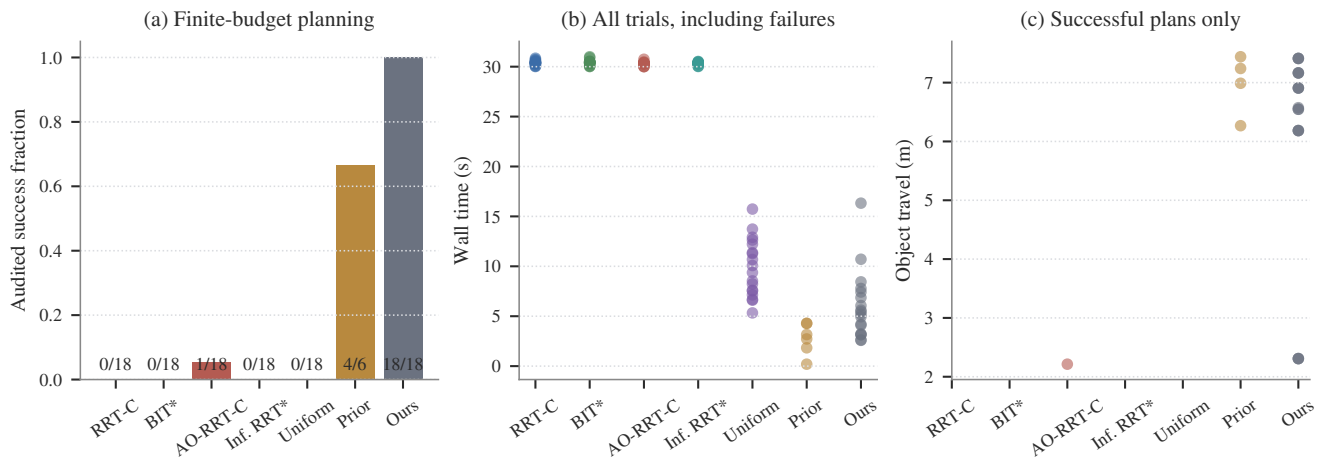


Fig. 3. External planner benchmark. Audited success and measured planning time include every completed trial; travel is conditional on successful planning. Uniform and scene-guided proposal regimes are distinguished. These finite-budget results characterize the tested implementation and initialization, not general planner superiority.